

## 3 Maximum likelihood estimation

### 3.1 Overview

So far we have been constructing estimators by informal reasoning. Maximum likelihood estimation (MLE) gives us a **systematic** way to construct estimators for any parametric family. Recall that the MLE is defined as follows:

**Definition 3.1.** Given data  $X_1, \dots, X_n$  drawn i.i.d. from a density  $f_\theta$ , where  $\theta$  is unknown, the **likelihood function** is

$$L(\theta; X_{1:n}) = \prod_{i=1}^n f_\theta(X_i).$$

The **log-likelihood function** is  $\ell(\theta; X_{1:n}) = \ln L(\theta; X_{1:n})$ . The **maximum likelihood estimator** (MLE) is the value of  $\theta$  that maximizes  $L$  (equivalently,  $\ell$ ):

$$\hat{\theta}_n = \arg \max_{\theta} \ell(\theta; X_{1:n}).$$

To construct the MLE and then analyze its statistical properties, we take the following steps:

1. Compute the (log-)likelihood function;
2. Compute the maximum likelihood estimator by maximizing it;
3. Derive the *exact* sampling distribution of the estimator;
4. Derive the mean-squared error (MSE) of the estimator and identify its statistical properties (biased? consistent?).

Step 1 requires fluency with algebraic manipulations, especially log and exponent properties. Step 2 requires fluency in derivative rules. Step 3 requires fluency with change-of-variables and sums and averages of iid random variables. As such, these problems synthesize a lot of the technical skills needed in future statistical theory courses. Furthermore, the steps are cumulative: you cannot do Step 3 correctly if you muck up Steps 1 and 2.

### 3.2 Worked examples

**Example 3.1** (Exponential distribution). Suppose  $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Exp}(\theta)$ , with density  $f_\theta(x) = \theta e^{-\theta x}$  for  $x > 0$ .

**Step 1: Log-likelihood.**

$$L(\theta; X_{1:n}) = \prod_{i=1}^n \theta e^{-\theta X_i} = \theta^n e^{-\theta \sum_{i=1}^n X_i} = \theta^n e^{-\theta S_n},$$

where  $S_n = X_1 + \dots + X_n$ . Taking the log:

$$\ell(\theta; X_{1:n}) = n \ln \theta - \theta S_n.$$

**Step 2: MLE.**

Setting the derivative to zero:

$$\ell'(\theta) = \frac{n}{\theta} - S_n = 0 \implies \hat{\theta}_n = \frac{n}{S_n} = \frac{1}{\bar{X}_n}.$$

Since  $\ell''(\theta) = -n/\theta^2 < 0$ , the log-likelihood is concave, confirming this is a maximum.

**Step 3: Sampling distribution.**

Since  $\text{Exp}(\theta) = \text{Gamma}(1, \theta)$ , by the MGF argument for sums of iid gammas,

$$\bar{X}_n \sim \text{Gamma}(n, n\theta).$$

Since  $\hat{\theta}_n = 1/\bar{X}_n$ , the sampling distribution of the MLE is the inverse gamma:

$$\hat{\theta}_n \sim \text{IG}(n, n\theta).$$

Read more about inverse gamma in the [distribution families](#) part of the website.

**Step 4: Bias, variance, and MSE.**

To compute moments, let  $Y = \bar{X}_n \sim \text{Gamma}(n, n\theta)$  and apply LOTUS with  $g(Y) = Y^{-k}$ :

$$\mathbb{E}[\hat{\theta}_n^k] = \int_0^\infty y^{-k} \cdot \frac{(n\theta)^n}{\Gamma(n)} y^{n-1} e^{-n\theta y} dy = \frac{(n\theta)^n}{\Gamma(n)} \cdot \frac{\Gamma(n-k)}{(n\theta)^{n-k}} = (n\theta)^k \frac{\Gamma(n-k)}{\Gamma(n)}.$$

Using  $\Gamma(n) = (n-1)\Gamma(n-1)$ :

$$\begin{aligned} \mathbb{E}[\hat{\theta}_n] &= n\theta \cdot \frac{\Gamma(n-1)}{\Gamma(n)} = \frac{n}{n-1} \theta, \\ \mathbb{E}[\hat{\theta}_n^2] &= (n\theta)^2 \cdot \frac{\Gamma(n-2)}{\Gamma(n)} = \frac{n^2}{(n-1)(n-2)} \theta^2. \end{aligned}$$

Therefore:

$$\text{bias}(\hat{\theta}_n) = \mathbb{E}[\hat{\theta}_n] - \theta = \frac{\theta}{n-1} > 0.$$

The estimator is *biased upward*, but the bias vanishes as  $n \rightarrow \infty$ .

$$\text{var}(\hat{\theta}_n) = \mathbb{E}[\hat{\theta}_n^2] - (\mathbb{E}[\hat{\theta}_n])^2 = \frac{n^2\theta^2}{(n-1)(n-2)} - \frac{n^2\theta^2}{(n-1)^2} = \frac{n^2\theta^2}{(n-1)^2(n-2)}.$$

Both  $\text{bias}^2(\hat{\theta}_n)$  and  $\text{var}(\hat{\theta}_n)$  go to zero as  $n \rightarrow \infty$ , so  $\text{MSE}(\hat{\theta}_n) \rightarrow 0$ : the estimator is **consistent**.

**Example 3.2** (Uniform distribution). Suppose  $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Unif}(0, \theta)$ , with density

$$f_\theta(x) = \begin{cases} \frac{1}{\theta}, & 0 < x < \theta \\ 0, & \text{otherwise.} \end{cases}$$

Before doing any calculations, let us visualize  $f_\theta(x)$  for two different values of  $\theta$ ; see Figure 30.

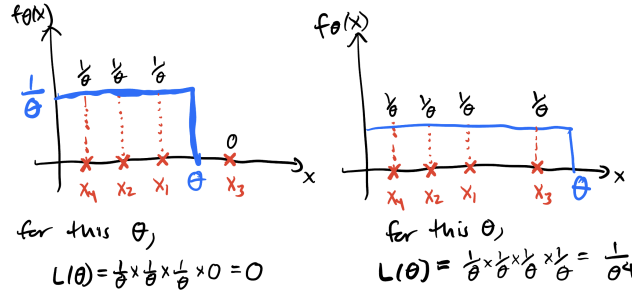


Figure 30: Density  $f_\theta(x)$  for two different values of  $\theta$ , with the data  $X_1, X_2, X_3, X_4$  marked on the  $x$ -axis.

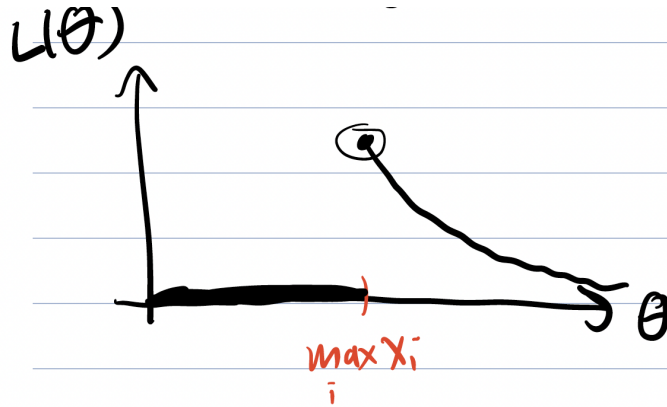
For both the left- and righthand plots, the data  $X_1, \dots, X_n$  stays the same: the red x's on the  $x$ -axis. Now, recall that the likelihood is the product  $L(\theta; X_{1:n}) = \prod_{i=1}^n f_\theta(X_i)$ . On the plots, this corresponds to the product of the heights of the vertical dotted lines. In the lefthand plot,  $\theta < X_3$ , so  $f_\theta(X_3) = 0$ , and the whole product is then zero, so  $L(\theta) = 0$ . This means the data could not have possibly been drawn from  $\text{Unif}(0, \theta)$  for the particular value of  $\theta$  in the lefthand plot. Indeed, if we *did* have  $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Unif}(0, \theta)$  for that  $\theta$ , then all the  $X_i$ 's would be forced to lie to the left of  $\theta$ . In the righthand plot, the value of  $\theta$  *does* contain all of the  $X_i$ , so the product of the  $f_\theta(X_i)$ 's is  $L(\theta) = (1/\theta)^n$ , with  $n = 4$ .

**Step 1: Likelihood** Let us now generalize the situation in the figure to arbitrary  $n$ . If  $\theta$  is smaller than any one of the  $X_i$ 's, then  $L(\theta) = 0$ . If  $\theta$  is larger than all of the  $X_i$ 's, then  $L(\theta) = 1/\theta^n$ . We can equivalently write this in the following form:

$$L(\theta; X_{1:n}) = \begin{cases} 0, & \theta < \max(X_1, \dots, X_n), \\ \frac{1}{\theta^n}, & \theta > \max(X_1, \dots, X_n). \end{cases} \quad (29)$$

**Step 2: MLE.** The function  $L(\theta)$  is plotted in Figure 31. We see by inspection that the largest value of  $L(\theta)$  is achieved at  $\theta = \max(X_1, \dots, X_n)$ . This must be the MLE. So

$$\hat{\theta}_n = \max(X_1, \dots, X_n).$$

Figure 31: The likelihood  $L(\theta)$  from (29), for the uniform example.**Step 3: Sampling distribution.**

The range of  $\hat{\theta}_n$  is  $[0, \theta]$ . For  $0 < t < \theta$ :

$$\begin{aligned} F_{\hat{\theta}_n}(t) &= P(\max(X_1, \dots, X_n) \leq t) = P(\{X_1 \leq t\} \cup \{X_2 \leq t\} \cup \dots \cup \{X_n \leq t\}) \\ &= P(X_1 \leq t)^n = \left(\frac{t}{\theta}\right)^n. \end{aligned}$$

Differentiating, the pdf of  $\hat{\theta}_n$  is:

$$f_{\hat{\theta}_n}(t) = \frac{n t^{n-1}}{\theta^n}, \quad 0 < t < \theta.$$

**Remark 3.1.** When  $\theta = 1$ , this pdf is a special case of the [Beta distribution family](#), which we will see next lecture when talking about Bayesian inference.

**Step 4: Bias, variance, and MSE.**

$$\mathbb{E}[\hat{\theta}_n] = \int_0^\theta t \cdot \frac{n t^{n-1}}{\theta^n} dt = \frac{n}{\theta^n} \cdot \frac{\theta^{n+1}}{n+1} = \frac{n}{n+1} \theta.$$

So  $\text{bias}(\hat{\theta}_n) = \frac{n}{n+1} \theta - \theta = -\frac{\theta}{n+1} < 0$ . The estimator is *biased downward* (which makes sense intuitively, since  $\max(X_i) \leq \theta$  always).

$$\mathbb{E}[\hat{\theta}_n^2] = \int_0^\theta t^2 \cdot \frac{n t^{n-1}}{\theta^n} dt = \frac{n}{\theta^n} \cdot \frac{\theta^{n+2}}{n+2} = \frac{n}{n+2} \theta^2.$$

$$\text{var}(\hat{\theta}_n) = \frac{n}{n+2} \theta^2 - \left(\frac{n}{n+1}\right)^2 \theta^2 = \frac{n \theta^2}{(n+1)^2(n+2)}.$$

Since both  $\text{bias}^2(\hat{\theta}_n) \rightarrow 0$  and  $\text{var}(\hat{\theta}_n) \rightarrow 0$  as  $n \rightarrow \infty$ , we have  $\text{MSE}(\hat{\theta}_n) \rightarrow 0$ : the estimator is **consistent**.